CalPolyPomona

College of Engineering

ECE 3301L.01

Lab 11

I^2C Bus Implementation

Randy Selner

Lance Taningco

Table #2

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Prof. Felix Pinai

California Polytechnic State University, Pomona

Department of Electrical and Computer Engineering

Objective

The purpose of this lab is to understand how to use the I2C (Inter-Integrated Circuit) protocol to connect a microcontroller with external devices. This lab focuses on using the I2C to interface with a Real-Time Clock (DS3231) and a digital thermometer (DS1621), allowing the PIC18F4620 microcontroller to send and receive data between devices. This lab also aims to teach us how to set up, initialize, and read data from these devices using C programming. In doing so, we hope to gain a better understanding of how data is transmitted through the SCL (clock) and SDA (data) pins. We will also utilize the logic analyzer to verify the correct data transmission between devices.

Summary

At the beginning of the lab, we copied the previous project from Lab 10 and prepared it for Lab 11, adding new I2C library files (`I2C.h, I2C_Soft.c, I2C_Support.h, I2C_Support.c`). This allows the microcontroller to communicate using the I2C protocol. In Part 1, we interfaced the DS1621 digital thermometer, learning how to write functions to initialize the device and read its temperature data in both Celsius and Fahrenheit. We then used the logic analyzer (Figure 11.2) to verify the communication by ensuring that the correct I2C address (0x48) responded with an acknowledge (ACK), while the second transaction responded with a not-acknowledge (NAK), indicating that talking to the wrong address results in such a result.

Part 2 involved the implementation of the DS3231 Real-Time Clock. It is meant to display the second, minute, hour, date, month, and year information through TeraTerm in real time, along with the current temperature reading. (Figure 11.3). This was accomplished by creating the `DS1621_Read_Temperature()` and `DS3231_Read_Time()` functions to fetch all registers and display them to TeraTerm.

In Part 3, we as students learned to manually set up the initial time values of the RTC by programming the device registers using the `DS3231_Setup_Time()` function. It is intended to pre-program all registers from 0x00 to 0x06 of the RTC with fixed numbers. In Part 4, we integrated the system with the remote control used in Lab 10 so that pressing a specific button (the ‘ + ’ button in this instance) could reset or reload the clock time, combining concepts from previous labs involving interrupts, displays, and sensor communication.

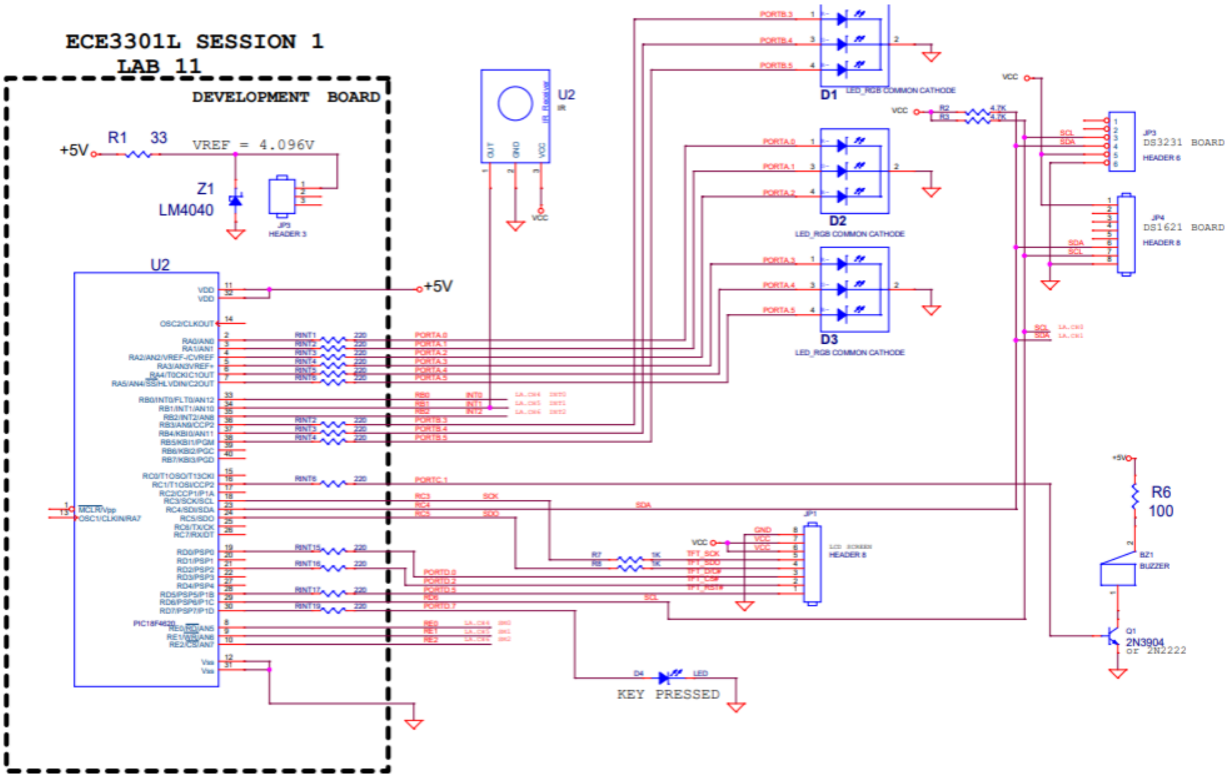

Figure 11.1: Schematic of circuit featuring the IR receiver, LEDs, LCD, KEY_PRESSED LED, buzzer, DS1621 thermometer, and DS3231 RTC

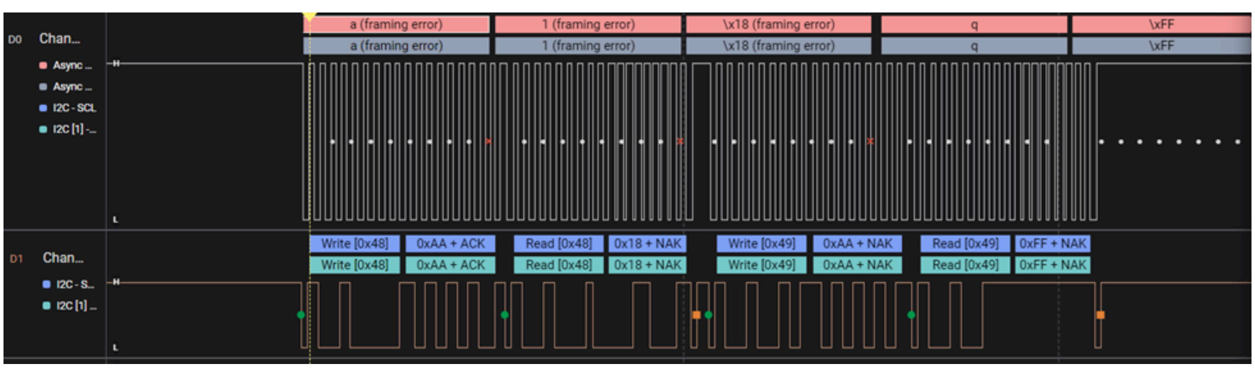

Figure 11.2: I2C transaction from `DS1621_Read_Temp()`

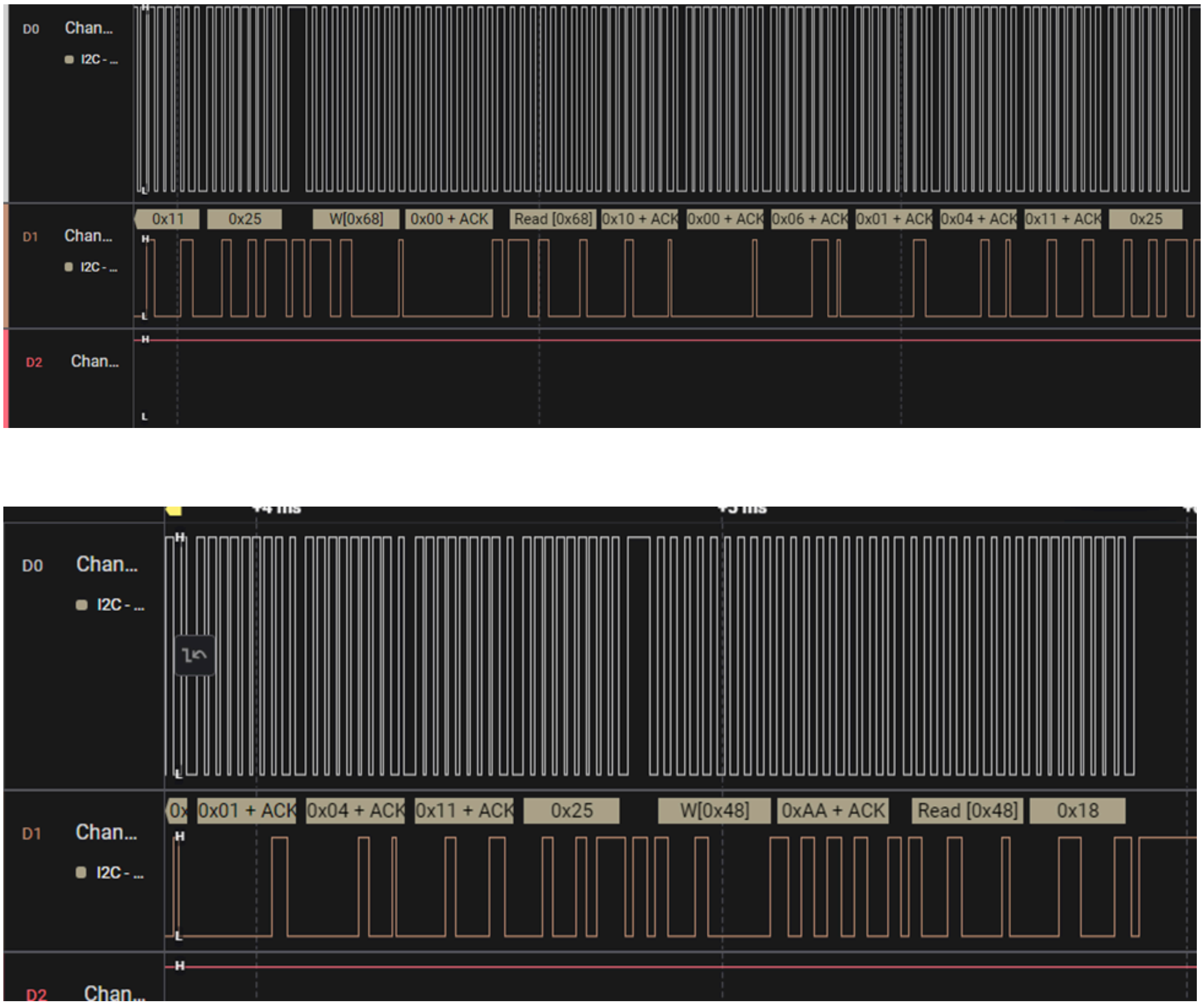

Figure 11.3: Logic analyzer time readout transaction

Conclusion

In conclusion, this lab showed us how the I2C protocol allows for multiple devices to communicate efficiently with a PIC18F microcontroller using only two data lines. By successfully being able to read and display the time and temperature data in TeraTerm, we developed a better understanding of how clocks and temperature sensors are integrated in real-world systems. Using the logic analyzer gave us a proper visual understanding of how data bits are transmitted and received between the PIC18F and the clock and sensor. Overall, this lab strengthened our foundation of embedded systems and showed the power of microcontroller-powered systems.